

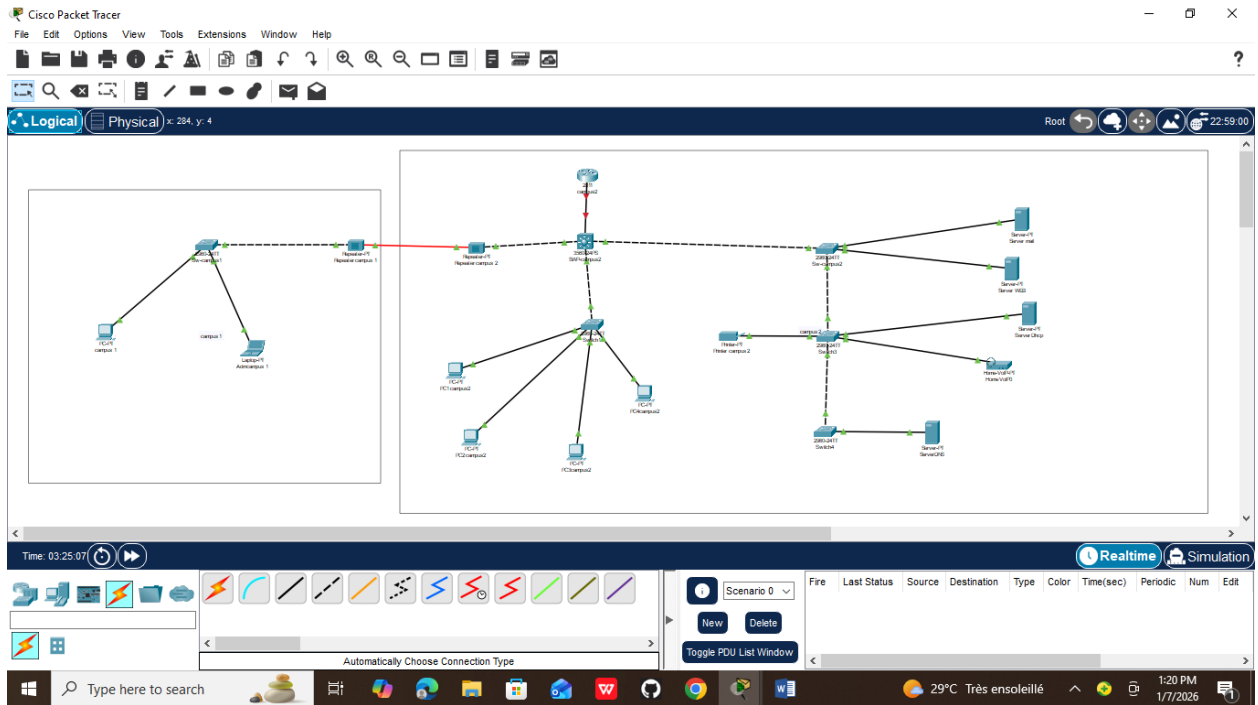
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TD9-RESAU1

FST

L3



La configuration afin de commencer

The image displays two screenshots of the Cisco Packet Tracer interface, showing network configuration for two switches.

Top Screenshot: Multilayer Switch0

The main window shows the configuration of Multilayer Switch0. The configuration commands entered are:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SWP-campus2
SWP-campus2(config)#vlan 10
SWP-campus2(config-vlan)#name ventes
SWP-campus2(config-vlan)#vlan 20
SWP-campus2(config-vlan)#marketing
% Invalid input detected at '^' marker.
SWP-campus2(config-vlan)#name marketing
SWP-campus2(config-vlan)#vlan 30
SWP-campus2(config-vlan)#name IT
SWP-campus2(config-vlan)#exit
SWP-campus2(config)#int g0/1
SWP-campus2(config-if)#switchport mode access
SWP-campus2(config-if)#int g0/2
SWP-campus2(config-if)#switchport mode access
SWP-campus2(config-if)#switchport mode trunk
SWP-campus2(config-if)#exit
SWP-campus2(config)#int f0/24
SWP-campus2(config-if)#switchport mode access
SWP-campus2(config-if)#switchport mode trunk
SWP-campus2(config-if)#exit
SWP-campus2(config)#%SPANTRIE-2-RECV_FVID_ERR: Received 802.1Q BPDU on non trunk
FastEthernet0/1 VLAN1.
%SPANTRIE-2-BLOCK_FVID_LOCAL: Blocking FastEthernet0/1 on VLAN0001. Inconsistent port
type.
```

The network diagram shows a central switch (SWP-campus2) connected to two repeaters (Repeatercampus1 and Repeatercampus2). The repeaters are connected to a 2960-24T switch (Switch0). The switch is connected to a PC (PC1campus1) and a laptop (Laptop-PT Admincampus 1). The switch is also connected to a PC (PC2campus2).

Bottom Screenshot: Switch2

The main window shows the configuration of Switch2. The configuration commands entered are:

```
SW-campus2>en
SW-campus2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW-campus2(config)#int f0/1
SW-campus2(config-if)#switchport mode trunk
SW-campus2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
SW-campus2(config-if)#exit
SW-campus2(config)#int range f0/4-9
SW-campus2(config-if-range)#switchport mode access
SW-campus2(config-if-range)#switchport access vlan 10
SW-campus2(config-if-range)#exit
SW-campus2(config)#int range f0/10-17
SW-campus2(config-if-range)#switchport mode access
SW-campus2(config-if-range)#switchport access vlan 20
SW-campus2(config-if-range)#exit
SW-campus2(config)#int range f0/18-24
SW-campus2(config-if-range)#switchport mode access
SW-campus2(config-if-range)#switchport access vlan 30
SW-campus2(config-if-range)#exit
SW-campus2(config)#
```

The network diagram shows a central switch (SW-campus2) connected to two repeaters (Repeatercampus1 and Repeatercampus2). The repeaters are connected to a 2960-24T switch (Switch0). The switch is connected to a PC (PC1campus1) and a laptop (Laptop-PT Admincampus 1). The switch is also connected to a PC (PC2campus2).

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 538, y: 143

Time: 02:43:37

Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
%SPANTRIE-2-RECV_FVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/4 VLAN1.
%SPANTRIE-2-BLOCK_FVID_LOCAL: Blocking FastEthernet0/4 on VLAN0001. Inconsistent port
type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

SWP-campus2>en
SWP-campus2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SWP-campus2(config)#vtp domain ius.com
Changing VTP domain name from NULL to ius.com
SWP-campus2(config)#vtp password 0000
Setting device VLAN database password to 0000
SWP-campus2(config)#vtp mode server
Device mode already VTP SERVER.
SWP-campus2(config)#vtp version 2
SWP-campus2(config)#spanning-tree mode rapid-pvst
SWP-campus2(config)#spanning-tree vlan 10 root primary
SWP-campus2(config)#spanning-tree vlan 20 root secondary
SWP-campus2(config)#int f0/2
SWP-campus2(config-if)#switchport port-security
Command rejected: FastEthernet0/2 is a dynamic port.
SWP-campus2(config-if)#switchport mode access
SWP-campus2(config-if)#switchport port-security
SWP-campus2(config-if)#switchport port-security maximum 3
SWP-campus2(config-if)#switchport port-security violation restrict
SWP-campus2(config-if)#exit
SWP-campus2(config)#exit
SWP-campus2#
```

Copy Paste

Root

Server-PT ServerMail

Server-PT ServerIweb

Server-PT Server2DHCP

Home-VoIP-PT Home VoIP0

Server-PT ServerDNS

Realtime Simulation

Type Color Time(sec) Periodic Num Edit

29°C Très ensoleillé 3:33 PM 12/30/2025

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 538, y: 143

Time: 02:44:46

Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Setting device VLAN database password to 0000
SWP-campus2(config)#vtp mode server
Device mode already VTP SERVER.
SWP-campus2(config)#vtp version 2
SWP-campus2(config)#spanning-tree mode rapid-pvst
SWP-campus2(config)#spanning-tree vlan 10 root primary
SWP-campus2(config)#spanning-tree vlan 20 root secondary
SWP-campus2(config)#int f0/2
SWP-campus2(config-if)#switchport port-security
Command rejected: FastEthernet0/2 is a dynamic port.
SWP-campus2(config-if)#switchport mode access
SWP-campus2(config-if)#switchport port-security
SWP-campus2(config-if)#switchport port-security maximum 3
SWP-campus2(config-if)#switchport port-security violation restrict
SWP-campus2(config-if)#exit
SWP-campus2(config)#exit
SWP-campus2#
%SYS-5-CONFIG_I: Configured from console by console

SWP-campus2#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/2, Fa0/3, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10 ventes	active	
20 marketing	active	
30 IT	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trinet-default	active	

SWP-campus2#

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Root

Server-PT ServerMail

Server-PT ServerIweb

Server-PT Server2DHCP

Home-VoIP-PT Home VoIP0

Server-PT ServerDNS

Realtime Simulation

Type Color Time(sec) Periodic Num Edit

29°C Très ensoleillé 3:33 PM 12/30/2025

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 538, y: 143

Time: 02:45:11

2960-24T Switch0 campus 1

Repeater-PT Repeatercampus1

Repeater-PT Repeatercampus2

PC-PT PC1campus1

Laptop-PT Admincampus 1

PC-PT PC2campus1

PC-PT PC2campus2

Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
1 default active Fa0/2, Fa0/3, Fa0/5, Fa0/6
Fa0/7, Fa0/8, Fa0/9, Fa0/10
Fa0/11, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/18
Fa0/19, Fa0/20, Fa0/21, Fa0/22
Fa0/23, Fa0/24, Gig0/1, Gig0/2

10 ventes active
20 marketing active
30 IT active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
SWP-campus2#show vlan

VLAN Name Status Ports
-----
1 default active Fa0/2, Fa0/3, Fa0/5, Fa0/6
Fa0/7, Fa0/8, Fa0/9, Fa0/10
Fa0/11, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/18
Fa0/19, Fa0/20, Fa0/21, Fa0/22
Fa0/23, Fa0/24, Gig0/1, Gig0/2

VLAN Type SAID MTU Parent RingNo BridgeNo Stp BrgdMode Transl Trans2
-----
1 enet 100001 1500 - - - - 0 0
10 enet 100010 1500 - - - - 0 0
20 enet 100020 1500 - - - - 0 0
--More--
```

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Server-PT ServerEmail

Server-PT ServerIweb

Server-PT Server2DHCP

Home-VoIP-PT Home VoIP0

Server-PT ServerDNS

Realtime Simulation

Type Color Time(sec) Periodic Num Edit

29°C Très ensoleillé 3:35 PM 12/30/2025

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 538, y: 143

Time: 02:48:43

2960-24T Switch0 campus 1

Repeater-PT Repeatercampus1

Repeater-PT Repeatercampus2

PC-PT PC1campus1

Laptop-PT Admincampus 1

PC-PT PC2campus1

PC-PT PC2campus2

Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Fa0/7, Fa0/8, Fa0/9, Fa0/10
Fa0/11, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/18
Fa0/19, Fa0/20, Fa0/21, Fa0/22
Fa0/23, Fa0/24, Gig0/1, Gig0/2

10 ventes active
20 marketing active
30 IT active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active

VLAN Type SAID MTU Parent RingNo BridgeNo Stp BrgdMode Transl Trans2
-----
1 enet 100001 1500 - - - - 0 0
10 enet 100010 1500 - - - - 0 0
20 enet 100020 1500 - - - - 0 0

SWP-campus2#show interface trunk
Port Mode Encapsulation Status Native vlan
Fa0/1 auto n-802.1q trunking 1
Fa0/4 auto n-802.1q trunking 1

Port Vlans allowed on trunk
Fa0/1 1-1005
Fa0/4 1-1005

Port Vlans allowed and active in management domain
Fa0/1 1,10,20,30
Fa0/4 1,10,20,30

Port Vlans in spanning tree forwarding state and not pruned
Fa0/1 1,10,20,30
Fa0/4 1,10,20,30

SWP-campus2#
```

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Server-PT ServerEmail

Server-PT ServerIweb

Server-PT Server2DHCP

Home-VoIP-PT Home VoIP0

Server-PT ServerDNS

Realtime Simulation

Type Color Time(sec) Periodic Num Edit

29°C Très ensoleillé 3:37 PM 12/30/2025

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 538, y: 143

2960-24T Switch0 campus 1 PC-PT campus1 Laptop-PT Admincampus 1 Repeater-PT Repeatercampus1 Repeater-PT Repeatercampus2 PC-PT PC1campus PC-PT PC2campus2

Time: 02:47:21

Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
SWP-campus2#show interface trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     auto      n-802.1q       trunking      1
Fa0/4     auto      n-802.1q       trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/4     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30
Fa0/4     1,10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30
Fa0/4     1,10,20,30

SWP-campus2#show vtp status
VTP Version capable : 1 to 2
VTP version running : 1
VTP Domain Name     : aus.com
VTP Pruning Mode     : Disabled
VTP Traps Generation: Disabled
Device ID            : 000D.B5E0.3200
Configuration last modified by 0.0.0.0 at 8-1-93 02:27:45
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
-----
VTP Operating Mode : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 8
Configuration Revision : 7
MD5 digest          : 0x68 0xA4 0xA6 0x01 0xEE 0x26 0x16 0x5D
                    : 0x14 0x33 0xD1 0xDC 0x5C 0x6C 0xB0 0x76

SWP-campus2#
```

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Root 09:38:30

Server-PT ServerMail

Server-PT ServerIweb

Server-PT Server2DHCP

Home-VoIP-PT Home VoIP0

Server-PT ServerDNS

Realtime Simulation

Type Color Time(sec) Periodic Num Edit

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 619, y: 50

2960-24T Switch0 campus 1 PC-PT campus1 Laptop-PT Admincampus 1 Repeater-PT Repeatercampus1 Repeater-PT Repeatercampus2 PC-PT PC1campus PC-PT PC2campus2

Time: 02:50:29

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
campus2 con0 is now available

Press RETURN to get started.

campus2>en
campus2#ping 192.168.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

campus2#
```

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Root 11:19:30

Server-PT ServerMail

Server-PT ServerIweb

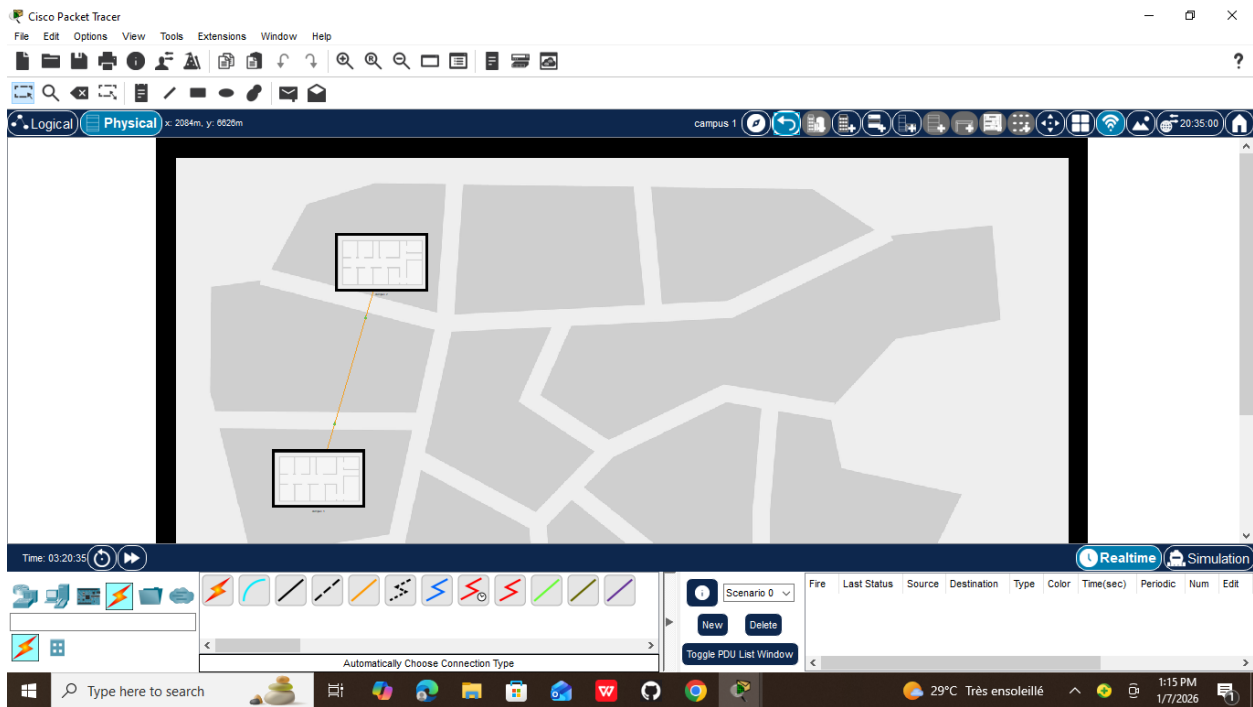
Server-PT Server2DHCP

Home-VoIP-PT Home VoIP0

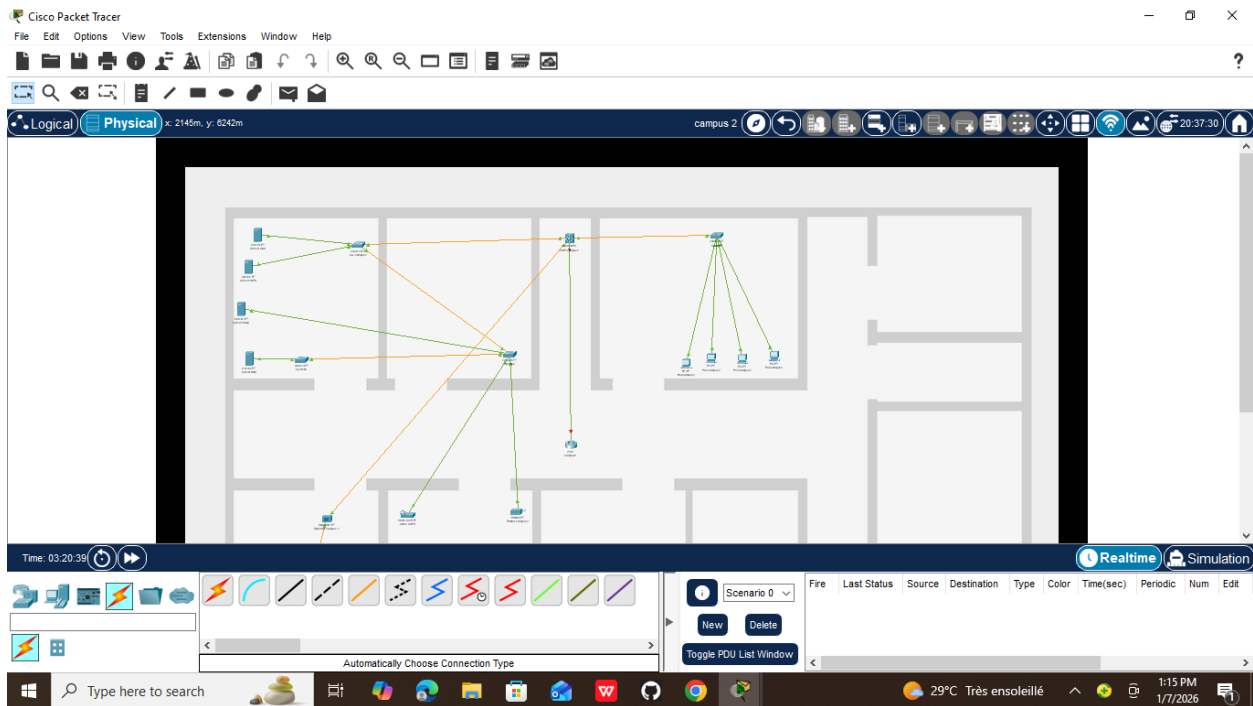
Server-PT ServerDNS

360 TOTAL SECURITY

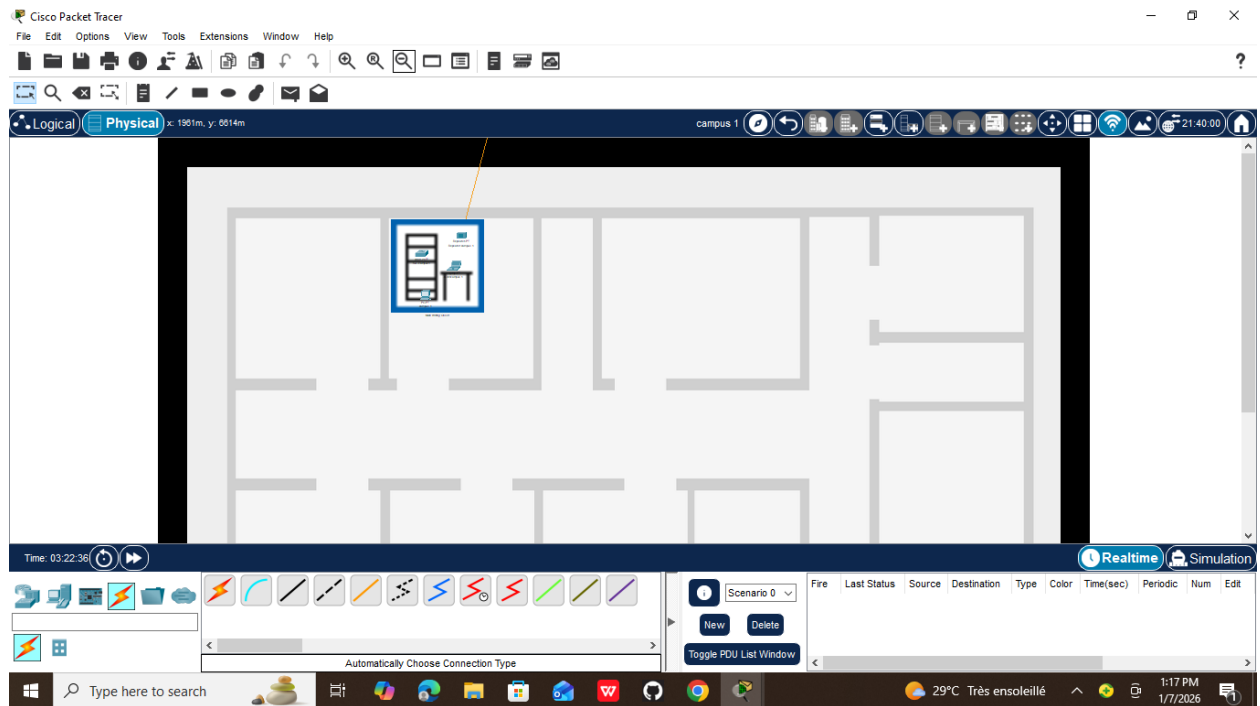
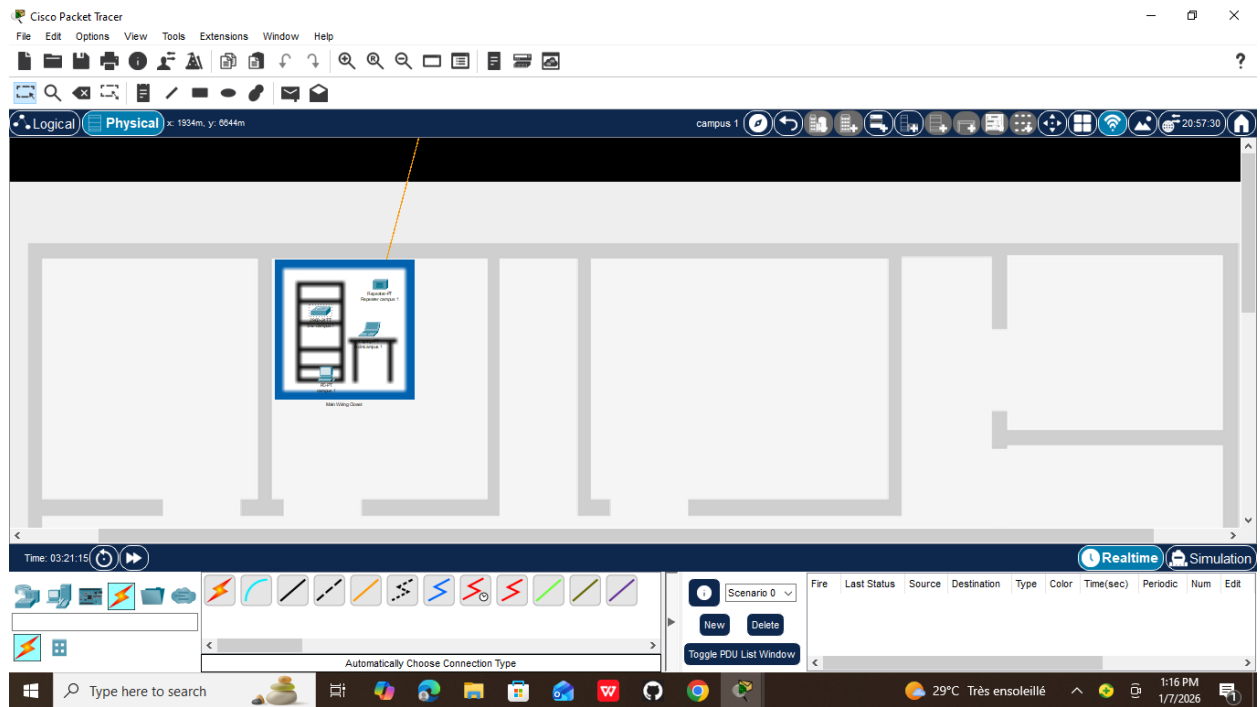
(D:) 105.3G/119.3G



Interconnectivite des campus 1 et 2



Représentation physique campus 2



Représentation physique du campus 1

En conclusion

Ce Td représente la fin du TD8 où on configurait d'abord les VLANs avant d'aller dans la partie physique et de gérer la représentation des deux campus physiquement afin de mieux comprendre le cours. J'ai appris à déplacer les appareils de manière physique afin que la connectivité soit plus réelle.